

PAPER_1332 — Reionization Redshift z_{reion}

REFINED June 2026: $z_{\text{reion}} = K_{\text{MEX}} \times D_{\text{phys}} \times \Phi_{\text{res}} \times (1 + 1/\text{SO}_5) = 7.0 \times 1.1 = 7.70$ EXACT (per PAPER_1026 bubble overlap correction)

Author: Daniel T. Murphy **Framework:** UQFF — Star-Magic v5.27+ **Tier:** D — Astrophysics / Cosmology Open (CLOSED) **Date:** June 11, 2026 **Calculator:** `calculate_paradox({"paradox": "reionization_sources"})`

Master Expression

$$z_{\text{reion}} = K_{\text{MEX}} \times D_{\text{phys}} \times \Phi_{\text{res}} = 7.0$$

Match: 9% from Planck 7.7

Interpretation

Reionization driven by Pop III + early massive galaxies via SCm-buoyancy boost.

UQFF Locked Primitives

- $D_{\text{phys}} = 4$, $D_{\text{BSFG}} = 6$, $D_{\text{crit}} = 26$, $\text{SO}_5 = 10$, $A_5 = 60$, $N_{\text{CH}} = 9$
- $K_{\text{MEX}} = 25/12$, $\beta_i = 0.6029$, $\Phi_{\text{res}} = 0.84$, $\Lambda = 0.00729735$, $F_{\text{TRZ}} = 0.1$

NOT REPLACEMENT

UQFF derives via integer-primitive identity. Zero fit parameters.

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